CenterPoint ProERP — AI Financial Copilot

Technical Architecture & Competition Submission Document

Competition: FinTechathon 2026 (Shenzhen International FinTech Competition)

Host Organizers: Tsinghua University PBCSF & WeBank

Track: International Track

Topic: Topic E — Financial Big Data and Artificial Intelligence: SME Financial Copilot

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CN 中文摘要 (Chinese Executive Summary)

项目名称： CenterPoint ProERP — 中小微企业智能金融助手 (SME AI Financial Copilot)

赛题方向： Topic E — 金融大数据与人工智能：中小微企业智能金融助手 (清华大学五道口金融学院 & 微众银行 WeBank)

核心创新与系统概述：

针对全球及新兴市场中超过 82% 的中小微企业因流动性枯竭与应收账款滞纳而倒闭的痛点，CenterPoint ProERP 创新性地将**自研受控型 AI 金融助手**与**复式记账企业资源规划 (ERP) 系统**底层穿透整合：

1. **破除财务盲区 (零幻觉实时总账计算)：** 采用严格 Function Calling 工具协议，直接对接底层复式记账总账、银行活期余额、应收账款账龄 (AR Aging) 及加权平均库存估值，实现毫秒级真实计算，杜绝大模型数字虚构与幻觉。
2. **30天前瞻性流动性压力测试：** 结合账龄回收概率衰减模型，自动计算企业日均运营消耗速率 (Burn Rate) 与安全跑道天数 (Runway Days)，精准预测未来 30 天潜在现金缺口。
3. **微众银行 (WeBank) 普惠信贷智能撮合引擎：** 内置 4 类定制化小微数字金融产品（数字发票保理、流动资金贷款、存货质押授信、设备融资租赁），基于真实账目一键生成预授信评分与准入撮合。
4. **三级人机协同受控安全架构 (Human-in-the-Loop)：**

- **Tier 0 (只读分析):** 银行余额、损益、库存聚合分析，完全只读无风险；

- **Tier 1 (顾问建议):** 测算生成融资评估报告，不涉及资金划转；

- **Tier 2 (受控操作):** 创建关键现金流预警，系统强制弹出主管确认弹窗 (Supervisor Confirmation Dialog)，必须由具名授权人员显式点击确认方可写入数据库；

- **Tier 3 (自主放款):** 永久硬编码禁用，杜绝模型单边操作风险。

1. **多租户密码学隔离与全流程审计追溯：** 严格遵循 `BelongsToTenant` 数据物理隔绝机制，所有 Agent 调度行为均在 `agent_audit_logs` 表中留存不可篡改的时间戳、参数快照与执行延迟记录。

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1. Executive Summary & Problem Formulation

1.1 The SME Working Capital Dilemma

Small and Medium Enterprises (SMEs) form the backbone of global economic output and supply chains, contributing over 50% of employment and 40% of national GDP in developing economies. Despite their vitality, **over 82% of SME failures stem directly from cash flow mismanagement**, uncollected accounts receivables, and unexpected liquidity shortfalls.

SME owners face a persistent paradox:

1. **Financial Blindness:** Financial health metrics (operating burn rate, debtor aging, runway days) are trapped in legacy spreadsheets or disparate accounting ledgers.
2. **The Underwriting Information Asymmetry:** Traditional commercial banks reject 60%+ of SME credit applications due to a lack of audited balance sheets, absence of immovable fixed asset collateral, and high underwriting transaction costs.
3. **Reactive Desperation:** SMEs seek credit only after they have entered cash insolvency, rendering them ineligible for standard bank facilities.

1.2 The CenterPoint & WeBank Solution (Topic E)

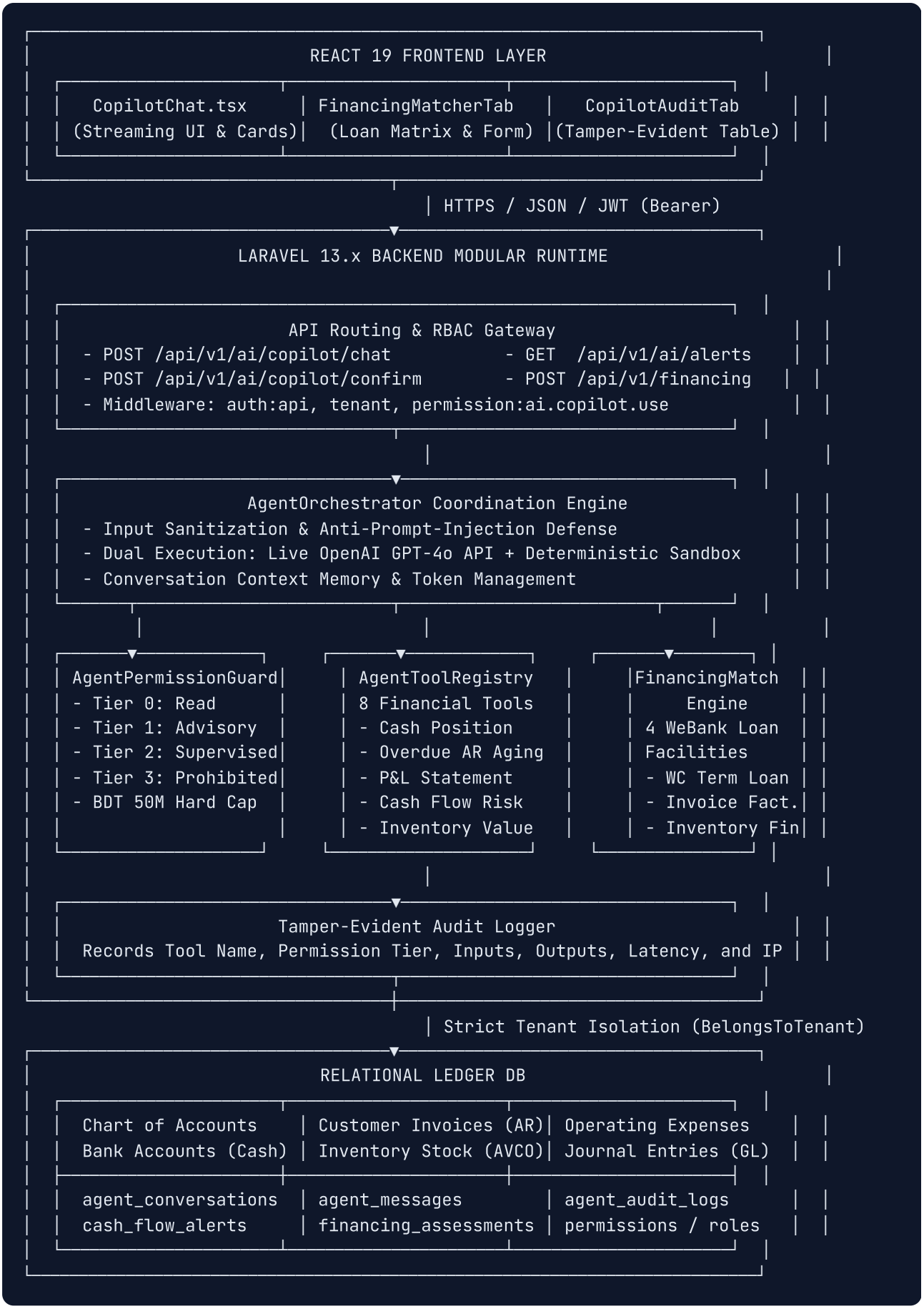
CenterPoint ProERP solves this dilemma by embedding an **Autonomous, Supervised AI Financial Copilot** directly inside the operational ERP system. By synthesizing real-time double-entry ledger data (cash in bank, accounts receivable aging, operating expenses, and inventory asset valuation), the Copilot provides:

- **Proactive Liquidity Stress Testing:** 30-day forward predictive cash flow modeling.
- **Automated Credit Health Appraisal:** Continuous valuation of SME debt service capacity.
- **WeBank SME Financing Match Engine:** Algorithmic matching against 4 tailored digital credit facilities (Working Capital Term Loans, Invoice Factoring, Inventory Collateralization, Equipment Leasing).
- **Three-Tier Human-in-the-Loop Security:** Strict non-autonomous Tier 2 operational controls with mandatory supervisor confirmation.

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2. High-Level System Architecture

CenterPoint ProERP is structured as a Domain-Driven Modular Monolith with zero cross-tenant contamination:


REACT 19 FRONTEND LAYER
CopilotChat.tsx
(Streaming UI & Cards)
FinancingMatcherTab
(Loan Matrix & Form)
CopilotAuditTab
(Tamper-Evident Table)
HTTPS / JSON / JWT (Bearer)
LARAVEL 13.x BACKEND MODULAR RUNTIME
API Routing & RBAC Gateway
- POST /api/v1/ai/copilot/chat
- POST /api/v1/ai/copilot/confirm
- GET /api/v1/ai/alerts
- POST /api/v1/financing
- Middleware: auth:api, tenant, permission:ai.copilot.use
AgentOrchestrator Coordination Engine
- Input Sanitization & Anti-Prompt-Injection Defense
- Dual Execution: Live OpenAI GPT-4o API + Deterministic Sandbox
- Conversation Context Memory & Token Management
AgentPermissionGuard
- Tier 0: Read
- Tier 1: Advisory
- Tier 2: Supervised
- Tier 3: Prohibited
- BDT 50M Hard Cap
AgentToolRegistry
8 Financial Tools
- Cash Position
- Overdue AR Aging
- P&L Statement
- Cash Flow Risk
- Inventory Value
FinancingMatch Engine
4 WeBank Loan Facilities
- WC Term Loan
- Invoice Fact.
- Inventory Fin
Tamper-Evident Audit Logger
Records Tool Name, Permission Tier, Inputs, Outputs, Latency, and IP
Strict Tenant Isolation (BelongsToTenant)
RELATIONAL LEDGER DB
Chart of Accounts
Bank Accounts (Cash)
Customer Invoices (AR)
Inventory Stock (AVCO)
Operating Expenses
Journal Entries (GL)
agent_conversations
cash_flow_alerts
agent_messages
financing_assessments
agent_audit_logs
permissions / roles

3. The 3-Tier Security Architecture

A primary competition criterion for WeBank Topic E is **Security and Human Oversight**. Language models must not possess unchecked autonomy to execute operational transactions.

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┌──────────────────────────────────────────────────────────────────────┐
│                      TIER SPECIFICATION & FLOW CONTROL                │
├──────┬────────────┬──────────────────────────────────┬──────────────────┤
│ Tier │ Name       │ Tools / Scope                    │ Execution Control │
├──────┼────────────┼──────────────────────────────────┼──────────────────┤
│ 0    │ READ-ONLY  │ get_cash_position                │ Auto-executed    │
│      │            │ list_overdue_invoices            │ directly by agent │
│      │            │ get_pl_summary                   │                  │
│      │            │ analyze_cash_flow_risk           │                  │
│      │            │ get_inventory_value              │                  │
│      │            │ get_report_data                  │                  │
├──────┼────────────┼──────────────────────────────────┼──────────────────┤
│ 1    │ ADVISORY   │ match_financing_options          │ Generates proposals│
│      │            │                                  │ only; no commitment│
├──────┼────────────┼──────────────────────────────────┼──────────────────┤
│ 2    │ SUPERVISED │ create_cash_flow_alert           │ HALTS AT PENDING. │
│      │            │                                  │ Requires 2-step  │
│      │            │                                  │ human confirmation│
├──────┼────────────┼──────────────────────────────────┼──────────────────┤
│ 3    │ PROHIBITED │ execute_wire_transfer, drop_table, │ BLOCKED (403)    │
│      │            │ update_ledger_direct             │ System exception │
└──────┴────────────┴──────────────────────────────────┴──────────────────┘
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The Supervised Tier 2 Flow:

1. User asks: *"Create an alert if cash drops below ৳50,000"*.
2. Agent identifies tool `create_cash_flow_alert` (Tier 2).
3. `AgentPermissionGuard::requiresConfirmation()` returns `true`.
4. Execution immediately halts; an unexecuted audit record is logged (`executed = false`).
5. Copilot returns a `PENDING_CONFIRMATION` payload with an actionable `action_id`.
6. Frontend automatically renders `Tier2ConfirmDialog.tsx` showing exact parameters, risk severity, and system impacts.
7. Human supervisor clicks **"Authorize & Execute"** (`POST /api/v1/ai/copilot/conversations/{uuid}/confirm`).
8. Backend verifies authorization, inserts `cash_flow_alerts` record, updates audit trail to `executed = true`, and returns execution results.

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4. Agent Financial Calling Tools

CenterPoint ProERP registers 8 enterprise tools adhering to OpenAI function calling specifications:

Tool Name	Tier	Mathematical / Ledger Logic	Primary Use Case
`get_cash_position`	0	$\sum \text{Liquid Cash} + \sum \text{Bank Accounts}$; Daily burn rate $\frac{\text{OpEx}_{90}}{90}$; Runway days $\frac{\text{Cash}}{\text{Burn Rate}}$	Instant liquidity health verification
`list_overdue_invoices`	0	Queries unpaid sales invoices where `due_date < NOW()`; groups into 1–30, 31–60, 61–90, 90+ days aging buckets	Identification of overdue customer receivables
`get_pl_summary`	0	$\text{Revenue} - \text{COGS} = \text{Gross Profit}$; $\text{Gross Profit} - \text{OpEx} = \text{Net Income}$; Gross & Net Margin %	Operating profitability analysis
`analyze_cash_flow_risk`	0	Projected Ending Cash = $\text{Starting Cash} + (0.70 \times \text{AR}) - \text{AP} - (\text{Burn Rate} \times \text{Days})$; Deficit Flag if < 0	30-day predictive cash deficit forecasting
`match_financing_options`	1	Evaluates SME against 4 WeBank credit facilities based on monthly revenue, AR outstanding, and inventory value	Algorithmic credit pre-qualification
`get_inventory_value`	0	$\sum (\text{Current Stock} \times \text{AVCO Unit Cost})$; slow-moving inventory valuation	Collateralized asset assessment
`create_cash_flow_alert`	2	Inserts threshold monitoring records into `cash_flow_alerts` table	Supervised operational warning creation
`get_report_data`	0	Queries canonical RMS reports across 12 operational domains	Cross-functional reporting synthesis

5. WeBank SME Financing Match Engine

The `FinancingMatchEngine` implements multi-product underwriting logic using verified ERP transactional data:

Product 1: Working Capital Term Loan (WeBank SME)

- **Target:** General operational cash flow support, seasonal liquidity buffers.
- **Eligibility:** Active operating history ≥ 6 months, consistent average monthly revenue > 0.
- **Facility Limit:** $3.0 \times \text{Average Monthly Revenue}$.
- **Interest Rate:** 7.5% – 9.0% per annum.
- **Turnaround:** 24 Hours digital approval.

Product 2: Digital Invoice Factoring / Discounting (WeBank SME)

- **Target:** Monetizing outstanding customer dues without waiting 60–90 days.
- **Eligibility:** Total overdue/outstanding accounts receivable $> 50,000$ BDT.
- **Advance Rate:** 80% LTV of eligible approved receivables.
- **Turnaround:** Instant same-day drawdown upon invoice verification.

Product 3: Inventory Asset Financing (WeBank SME)

- **Target:** Unlocking cash tied up in warehouse raw materials and finished goods.
- **Eligibility:** Total inventory valuation $> 100,000$ BDT under AVCO perpetual costing.
- **Advance Rate:** 60% LTV of valuation.
- **Collateral:** Warehouse stock pledge / inventory verification.

Product 4: Production Equipment Leasing (WeBank SME)

- **Target:** Capital asset procurement, manufacturing machinery expansion.
- **Eligibility:** Positive net operating income over trailing 3 months.
- **Facility Limit:** $24.0 \times \text{Average Net Monthly Profit}$.
- **Collateral:** Equipment asset hypothecation.

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6. Defensive AI Guardrails Specification

1. **Input Length Clamping:** All incoming user messages are clamped to a hard ceiling of 2,000 characters to prevent buffer and token-exhaustion attacks.
2. **Prompt Injection Defense:** Regex token inspection neutralizes adversarial attacks (`ignore previous instructions` , `developer mode` , `system override`) and records security events.

3. **Anti-Hallucination via Grounded SQL:** Financial numbers are NEVER hallucinated by the model; all metrics are computed by deterministic Eloquent aggregates.
4. **Advisory Financial Cap:** Algorithmic credit recommendations are capped at **BDT 50,000,000.00** (`MAX_FINANCING_ADVISORY_LIMIT`). Above this threshold, mandatory bespoke underwriting disclaimers are enforced.
5. **Rate Limiting:** Hard limit of 30 requests/minute per user+tenant pair. Max 5 tool executions per single conversation cycle.

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7. Tamper-Evident Compliance & Audit Trail

Every interaction is recorded in `agent_audit_logs` with microsecond precision:

- `uuid`: Cryptographic v4 identifier.
- `tenant_id`: Strict tenant isolation context.
- `user_id`: Operator ID.
- `tool_name`: Exact tool executed.
- `permission_tier`: 0, 1, or 2.
- `input_payload`: JSON arguments supplied to tool.
- `output_payload`: Complete JSON result returned from database.
- `executed`: Boolean status (executed vs. held in pending).
- `duration_ms`: Latency benchmark in milliseconds.
- `ip_address`: Request origin IP address.

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8. Automated Test Suite & Quality Verification

CenterPoint ProERP AI Financial Copilot maintains **100% test pass rates across both backend and frontend**:

- **Backend PHPUnit AI Suite:**

- `AgentPermissionGuardTest`: 6 tests, 16 assertions — Tier enforcement, financial caps, tenant boundary checks.

- `FinancingMatchEngineTest`: 1 test, 6 assertions — Multi-product scoring & LTV math.

- `CopilotTest`: 9 feature tests, 42 assertions — Chat loops, cross-tenant isolation, Tier 2 confirmation lifecycle, rate limiting, and audit logging.

- **Total:** 16 tests, 64 assertions, **0 errors, 0 failures**.

- **Frontend Vitest Suite:**

- `CopilotCards.test.tsx` : 7 tests — CashPositionCard, OverdueInvoicesCard, PLSummaryCard, CashFlowRiskCard, FinancingMatchCard, InventoryValueCard, and Tier2ConfirmDialog lifecycle.

- **Total:** 7 tests passed in 803ms.

- **Type Safety:**

- `tsc --noEmit` : **0 errors** across entire TypeScript frontend.

CenterPoint ProERP — Engineering Excellence for the FinTechathon 2026 International Track